

Validity-Aware Spectral Diagnostics for Multifractal Random Walk Inference Under Finite Samples

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Abstract

Multiscale fluctuations in complex stochastic systems often exhibit heavy tails, intermittency, and nonlinear scaling spectra, features that are commonly interpreted as evidence of multifractality. Under finite observations, however, this interpretation is intrinsically ambiguous: stable monofractal processes may produce apparently regular scaling, heavy-tailed or regime-switching signals may generate spurious curvature, and high-order empirical moments can be strongly affected by sample length, scale range, and the choice of q -grid. This paper proposes CMIN-SR, a validity-aware spectral diagnostic framework for assessing multifractal random walk (MRW)-like evidence under finite samples. Rather than treating neural inference as unconstrained spectrum regression, CMIN-SR separates empirical spectrum estimation from spectral interpretation. An observed scaling spectrum is projected onto competing monofractal and MRW spectral families, and calibrated diagnostic scores are reported for scaling stability, nonlinear curvature, monofractal compatibility, MRW compatibility, boundary behavior, and instability.

Controlled analytic-spectrum experiments show that the spectral-geometry calibrator learns the intended distinction between linear, boundary, and curved spectra. Across three random seeds, linear monofractal spectra receive low MRW compatibility and low curvature scores, whereas strongly curved MRW spectra receive high curvature and MRW-compatibility scores. In representative runs, the normalized MRW-compatibility score increases from about 0.13 for linear spectra to about 0.94 for curved MRW spectra. On raw finite samples, zeta-alignment reduces monofractal false positives in the documented $T = 1024$ comparison, lowering fGn MRW compatibility from about 0.73 to 0.36. However, deterministic and classical empirical-spectrum estimators, including aggregate structure functions, MFDFA, and lightweight Haar wavelet-style controls, do not remove the short-window curvature bottleneck: MFDFA improves some error and high- λ^2 detection metrics, but projected λ^2 remains weakly and inconsistently correlated with the known intermittency parameter in the tested regime. These findings identify empirical spectrum estimation and finite-sample curvature recovery, rather than spectral interpretation alone, as the central bottleneck in short-window MRW inference. CMIN-SR should therefore be understood as a calibrated diagnostic framework for multifractal spectral evidence, not as proof of an MRW data-generating mechanism.

1 Introduction

Many complex stochastic systems exhibit fluctuations that are neither independent nor characterized by a single scale. Financial returns, turbulent cascade surrogates, and other intermittent signals often display heavy-tailed increments, clustered activity, long-range dependence in amplitudes, and nonlinear scaling of empirical moments. Multifractal analysis provides a compact language for such behavior by describing how moments of different orders scale across observation scales. In this language, a process is not summarized only by a single Hurst or roughness exponent, but by a scaling function $\zeta(q)$ and, in the broader formalism, by an associated singularity spectrum $f(\alpha)$

[16, 11]. This viewpoint has been widely used in complex systems and econophysics, where apparent multiscaling is often interpreted as a signature of cascade-like organization or intermittent volatility dynamics [12, 23, 10, 14].

Among the models used to represent such spectra, the multifractal random walk (MRW) is particularly attractive because it gives a low-dimensional analytic description of nonlinear scaling. In the convention used in this work, the MRW-like spectrum is written as

$$\zeta_{\text{MRW}}(q) = qH - \frac{1}{2}\lambda^2 q(q-2),$$

where H controls the leading monofractal scaling and λ^2 controls the curvature or intermittency of the spectrum [3, 5]. When λ^2 is close to zero, the spectrum is nearly linear and resembles a monofractal scaling law such as that of fractional Brownian motion or fractional Gaussian noise [24]. When λ^2 is larger, the spectrum becomes visibly curved and is more compatible with an MRW-like interpretation. This simple analytic form makes MRW a useful reference model for finite-sample spectral diagnostics.

The central difficulty is that empirical multifractal spectra are not observed directly. They are estimated from finite time series through structure functions, detrended fluctuation statistics, wavelet-transform modulus maxima, wavelet leaders, or related multiscale estimators [25, 26, 1, 19, 18, 28]. These estimators are sensitive to the available sample length, the scale range, the chosen q -grid, and the stability of high-order moments. A finite monofractal sample can generate apparent curvature; heavy-tailed data can destabilize high- q moments; and regime-switching or volatility-clustering signals can produce scaling patterns that resemble multifractal behavior without implying an MRW mechanism [8, 21, 15, 29, 6]. Thus the practical problem is not merely to fit a curved spectrum, but to decide whether the curvature is stable, whether a monofractal explanation is already sufficient, and whether the finite sample contains enough information to support an MRW-like interpretation.

Traditional multifractal estimators are indispensable, but they typically leave this decision to post-hoc interpretation. A structure-function regression may produce an empirical $\zeta(q)$, and MF DFA or wavelet-based methods may produce more robust multiscale estimates, but the analyst must still decide whether the observed curvature is physical, finite-size induced, tail-driven, or a consequence of scale selection. Even for wavelet-based multifractal procedures, reliable curvature estimation depends on the asymptotic regime and the available scale support [22, 4]. Conversely, unconstrained neural networks can be trained to regress parameters or spectra from finite signals, but a black-box prediction does not by itself enforce spectral geometry, distinguish monofractal and MRW explanations, or expose when the inverse problem is under-identified. This is precisely the setting where constrained scientific machine learning is useful: data-driven components should be combined with mathematical structure, physically interpretable coordinates, and explicit diagnostics rather than used as unrestricted replacements for the model [2, 13, 27, 20]. This places the neural component of CMIN-SR closer to data-driven inverse modeling and simulation-based diagnostic inference than to unrestricted time-series prediction. Related scientific machine-learning work makes a similar distinction between using neural networks as unconstrained predictors and using them as structure-aware components in physical modeling pipelines [9, 20]. The term simulation-based is used here in this broad methodological sense; the present paper does not perform posterior inference or claim a full likelihood-free Bayesian analysis.

This paper introduces CMIN-SR, a validity-aware spectral diagnostic framework for finite-sample MRW inference. The aim is not to claim guaranteed recovery of the true MRW intermittency parameter from short observations. Instead, the framework separates the problem into three interpretable stages. First, a finite signal is mapped to an empirical scaling spectrum $\zeta_{\text{emp}}(q)$. Second,

this empirical spectrum is projected onto two competing analytic families: a linear monofractal spectrum and a parabolic MRW spectrum. Third, the resulting residuals, curvature diagnostics, boundary scores, and instability indicators are passed to a spectral-geometry calibrator that reports diagnostic scores for scaling stability, nonlinear curvature, monofractal compatibility, MRW compatibility, and low-curvature boundary behavior. In this formulation, λ_{proj}^2 is a projection coordinate within the MRW family, not proof that the observed signal was generated by an MRW process.

This diagnostic formulation leads to a different scientific question from ordinary parameter regression. Rather than asking only whether a neural model can output a number close to λ^2 , we ask whether the finite empirical spectrum contains stable evidence that favors a curved MRW explanation over a linear monofractal one. This distinction is important because stable scaling alone is not sufficient evidence for multifractality. It is also important because failure can occur at different levels: the analytic spectral geometry may be interpretable, while the raw empirical spectrum may be too noisy or too short to preserve the relevant curvature.

The contributions of this work are as follows.

1. We formulate finite-sample MRW inference as a validity-aware spectral diagnostic problem. The framework explicitly separates empirical spectrum estimation, monofractal/MRW projection, residual geometry, curvature diagnostics, boundary behavior, and instability warnings.
2. We introduce and evaluate a spectrum-space calibration layer that operates on analytic and semi-analytic $\zeta(q)$ curves. In clean spectrum space, the calibrator separates linear monofractal spectra, low-curvature boundary MRW spectra, strongly curved MRW spectra, and unstable distorted spectra, demonstrating that the spectral interpretation layer is learnable when the input geometry is controlled.
3. We provide a finite-sample identifiability analysis showing that the main bottleneck lies upstream, in empirical spectrum estimation. Under the tested short-window settings, both deterministic structure-function estimators and neural empirical-spectrum estimators struggle to recover MRW curvature reliably. This negative result is central to the paper: it motivates the interpretation of CMIN-SR as a calibrated diagnostic framework rather than a guaranteed short-window MRW parameter recovery engine.

The remainder of the paper is organized as follows. Section 2 introduces the finite-sample spectral inference problem, the monofractal and MRW projection families, and the diagnostic inverse formulation. Section 3 describes the CMIN-SR framework and shows how MRW samples are simulated. Section 4 presents the experimental design, including analytic spectra, synthetic process families, classical baselines, zeta-space noise, failure attribution, and a cautious real-data sanity check. Section 5 reports the main results. Section 6 analyzes failure modes, and Section 7 discusses interpretation and limitations.

2 Background and Problem Formulation

2.1 Finite-sample scaling spectra

Let x_t , $t = 1, \dots, T$, denote a finite sequence of observed increments. In the synthetic experiments below, x_t is a standardized return or increment sequence rather than an already integrated path. A common route to multifractal analysis is to examine how empirical moments vary across aggregation

scales. For a scale a , define the aggregated increment

$$\Delta_a X_t = \sum_{i=0}^{a-1} x_{t+i}. \quad (1)$$

A structure-function estimate is then based on

$$S_q(a) = \frac{1}{T-a+1} \sum_{t=1}^{T-a+1} |\Delta_a X_t|^q, \quad (2)$$

and, over a finite scale set $\mathcal{A}$, the scaling exponent is estimated from the approximate relation

$$S_q(a) \approx C_q a^{\zeta(q)}. \quad (3)$$

In practice, $\zeta(q)$ is obtained by regressing $\log S_q(a)$ on $\log a$ for each q . The deterministic estimators used in the present study include ordinary least-squares regression, trimmed regression over scales, bootstrap resampling of scale subsets, and a lightly smoothed $\zeta(q)$ curve. These choices are deliberately simple: they allow us to ask whether the curvature signal is identifiable before attributing failure to a particular neural architecture.

The finite-sample setting is crucial. The scale set is short, the q -grid is finite, and high-order moments are sensitive to rare events. In the main experiments the default grid is

$$q \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\},$$

and the default scale set is drawn from powers of two such as $\{2, 4, 8, 16, 32, 64\}$, subject to the sample length. These choices are not claimed to be universal. They are the tested regime in which CMIN-SR is evaluated.

2.2 From scaling exponents to singularity spectra

The scaling function $\zeta(q)$ is one representation of a multifractal spectrum. In the classical multifractal formalism, a singularity spectrum can also be obtained through a Legendre-type transformation,

$$\alpha(q) = \frac{d\zeta(q)}{dq}, \quad f(\alpha(q)) = q\alpha(q) - \zeta(q) + 1, \quad (4)$$

for one-dimensional signals under the usual formal assumptions [16, 11]. This paper does not claim a validated empirical reconstruction of $f(\alpha)$ for arbitrary real data. Instead, $f(\alpha)$ is used to connect the analytic MRW projection family with standard multifractal geometry. The reported empirical diagnostics focus on $\zeta(q)$, projection residuals, curvature, and stability.

2.3 Monofractal and MRW projection families

A monofractal spectrum is represented by the linear family

$$\zeta_{\text{mono}}(q) = qH, \quad (5)$$

where H is a scaling exponent. This family is an essential competitor: stable scaling by itself is not evidence of multifractality. Fractional Gaussian noise, for example, can exhibit stable monofractal scaling while having no nonlinear curvature in $\zeta(q)$ [24].

The MRW reference family used here follows the parabolic convention

$$\zeta_{\text{MRW}}(q) = qH - \frac{1}{2}\lambda^2 q(q-2), \quad (6)$$

where H controls the leading linear component and λ^2 controls spectral curvature [3, 5]. The phrase “MRW-compatible” in this paper always refers to compatibility with this projection family under the selected q -grid and residual criterion. The projected coordinate λ_{proj}^2 is not evidence that the observed time series was generated by an MRW mechanism. Other MRW parameterizations exist; all reported curvature coordinates in this paper use the convention in Eq. (6).

2.4 Projection residuals as diagnostic quantities

Given an empirical spectrum $\hat{\zeta}(q)$, CMIN-SR computes two least-squares projection coordinates: one for the monofractal family and one for the MRW family. In schematic form,

$$\hat{H}_{\text{mono}} = \arg \min_H \sum_{q \in \mathcal{Q}} \left[\hat{\zeta}(q) - qH \right]^2, \quad (7)$$

and

$$(\hat{H}_{\text{proj}}, \hat{\lambda}_{\text{proj}}^2) = \arg \min_{H, \lambda^2} \sum_{q \in \mathcal{Q}} \left[\hat{\zeta}(q) - qH + \frac{1}{2}\lambda^2 q(q-2) \right]^2. \quad (8)$$

The implementation clamps the MRW curvature coordinate to a nonnegative admissible range. The corresponding residual norms are used together with curvature diagnostics. A low MRW residual is not sufficient by itself; the monofractal residual must also be considered. If the two residuals are nearly equal and λ_{proj}^2 is small, the sample is better described as a boundary case rather than as a strong MRW case. Here, “boundary” is an operational diagnostic label for spectra that lie near the monofractal/MRW projection boundary in the tested finite-sample geometry; it is not introduced as a separate theoretical process class.

2.5 The diagnostic inverse problem

The finite-sample inverse problem studied here is therefore

$$x(t) \longrightarrow \hat{\zeta}(q) \longrightarrow \{\hat{H}_{\text{mono}}, \hat{H}_{\text{proj}}, \hat{\lambda}_{\text{proj}}^2, r_{\text{mono}}, r_{\text{MRW}}\} \longrightarrow \text{diagnostic scores}. \quad (9)$$

The purpose is not to recover a latent parameter at all costs. The purpose is to decide whether the finite empirical spectrum gives stable evidence for a linear monofractal explanation, a low-curvature boundary explanation, a curved MRW-family explanation, or an unstable/tail-driven pattern.

3 CMIN-SR Diagnostic Framework

3.1 Overview

CMIN-SR is designed as a constrained diagnostic pipeline rather than as a free-form spectrum regressor. The pipeline separates two tasks that are often mixed together. The first task is empirical spectrum estimation: from a finite signal, obtain a usable $\hat{\zeta}(q)$. The second task is spectral interpretation: decide how this empirical spectrum relates to monofractal and MRW-family explanations. This separation matters because a clean spectrum and a noisy spectrum can lead to very different conclusions even when the downstream interpretation rule is the same.

The framework reports diagnostic scores rather than mechanism probabilities. The main scores are:

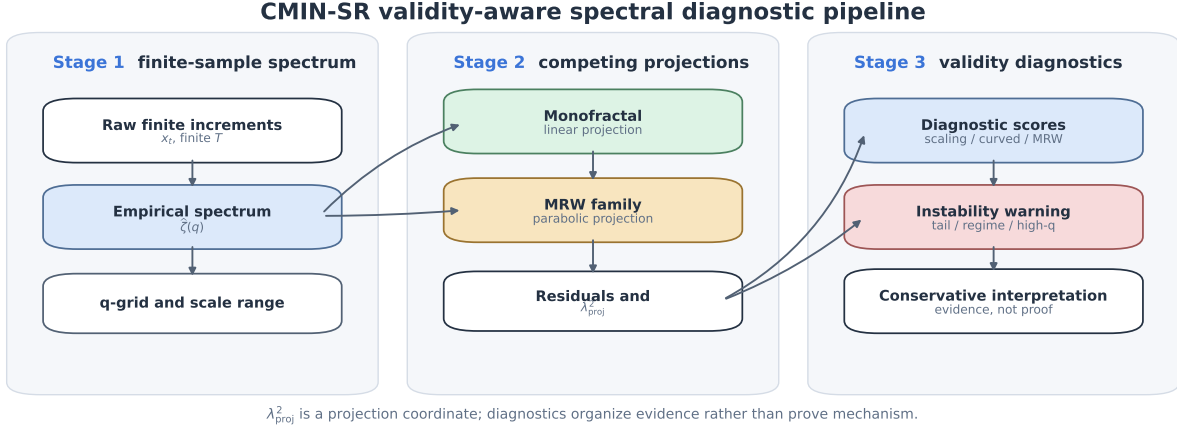


Figure 1: CMIN-SR diagnostic pipeline. Raw finite signals are mapped to empirical spectra, projected onto monofractal and MRW spectral families, and interpreted through residual geometry, curvature, boundary, and instability diagnostics. The output is organized diagnostic evidence, not proof of an MRW data-generating mechanism.

- p_{scaling} : evidence for stable scaling over the tested scales;
- p_{curved} : evidence for nonlinear curvature beyond a linear monofractal spectrum;
- p_{mono} : compatibility with a monofractal explanation;
- p_{MRW} : compatibility with a curved MRW-family projection;
- p_{boundary} : evidence that the spectrum lies near the monofractal/MRW boundary.

These scores should be read jointly with residuals and instability warnings. For example, high p_{scaling} alone is not evidence of multifractality, and high MRW compatibility should not be interpreted without checking the monofractal residual. The notation p denotes a normalized diagnostic score produced by the model; it should not be read as a statistically validated posterior probability.

3.2 How MRW samples are simulated

The synthetic MRW samples used in the project are generated in a way that matches the analytic projection family. A log-volatility modulation ω_t is sampled as a stationary Gaussian field with logarithmic covariance,

$$\text{Cov}(\omega_t, \omega_{t+k}) = \lambda^2 \log \frac{L}{k + \epsilon} \quad \text{for } k < L, \quad (10)$$

and zero covariance beyond the integral scale L . The implementation samples this Gaussian field using a circulant embedding of the Toeplitz covariance matrix. The field is shifted by $-\text{Var}(\omega_t)/2$, which keeps the lognormal modulation centered.

For the fGn-based MRW variant used in the CMIN-SR process generator, fractional Gaussian noise $\epsilon_t^{(H)}$ is sampled with Hurst exponent H , and the increment sequence is

$$\Delta X_t = \sigma \exp(\omega_t) \epsilon_t^{(H)}. \quad (11)$$

The path X_t is obtained by cumulative summation. In most diagnostic experiments the model receives the standardized increments ΔX_t , not the integrated path. The theoretical scaling curve

MRW simulation example ($H = 0.60, \lambda^2 = 0.12$)

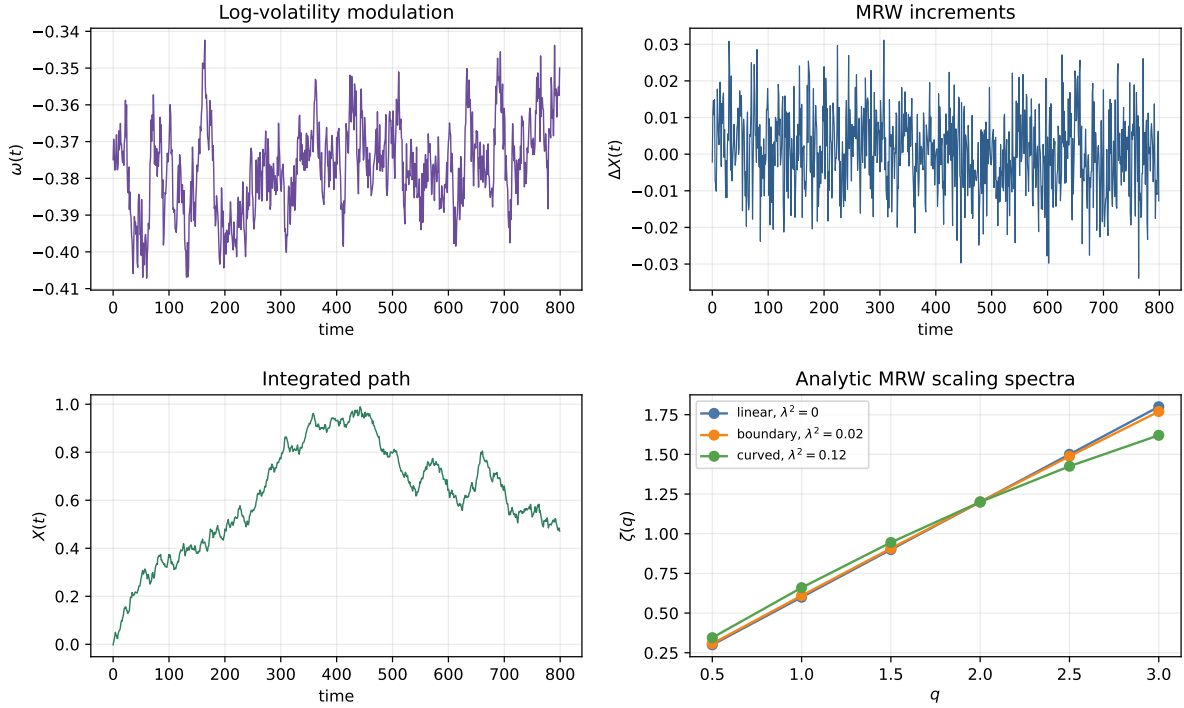


Figure 2: Example MRW simulation used for controlled experiments. The log-volatility modulation creates intermittent increments, while the analytic spectrum shows how increasing λ^2 bends the otherwise linear scaling law. The data used to generate this figure are saved with the paper assets for reproducibility.

attached to these samples is Eq. (6). Low- λ^2 MRW samples therefore act as boundary cases: their spectra are close to linear even though they are generated from the MRW family.

3.3 Analytic spectrum-space calibration

To isolate the interpretation problem from the raw estimation problem, the project first trains and evaluates a small spectral-geometry calibrator on analytic and semi-analytic $\zeta(q)$ curves. The analytic dataset includes: linear monofractal spectra, low-curvature boundary MRW spectra, strongly curved MRW spectra, noisy monofractal spectra, noisy MRW spectra, heavy-tail distorted spectra, regime-like apparent spectra, and ambiguous mild-curvature spectra. Each sample contains the input $\zeta(q)$, the monofractal and MRW projection curves, residual norms, MRW-vs-monofractal gain, curvature score, linearity score, boundary score, and instability targets.

The calibrator itself is intentionally small: it is a multilayer perceptron over spectrum-space features, not a large raw time-series backbone. Its role is to learn the geometry of the diagnostic space. This design allows us to ask a clean question: if the spectrum is correct, can the diagnostic layer separate linear, boundary, curved, and unstable cases? The answer is positive in the analytic experiments, which motivates the later failure analysis on raw finite samples.

Analytic Monofractal and MRW Spectrum Geometry

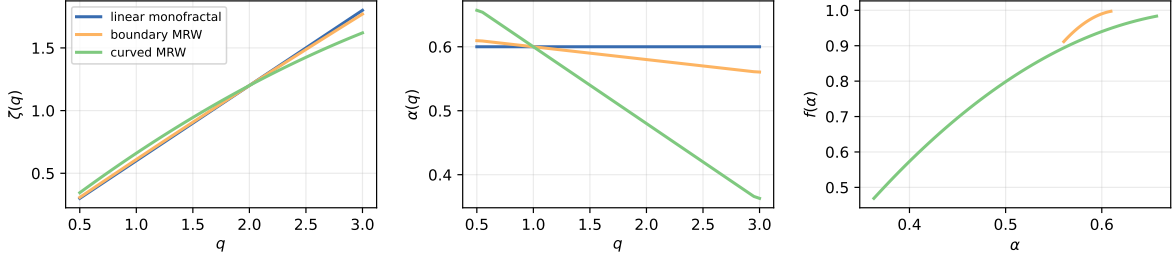


Figure 3: Analytic monofractal and MRW spectrum geometry. The figure is not an empirical reconstruction result; it visualizes the analytic decoder used to interpret monofractal, boundary, and curved MRW-family spectra.

3.4 Raw zeta alignment and curvature preservation

The project also studies whether a raw encoder can learn a cleaner $\hat{\zeta}(q)$ from finite time series. Raw zeta alignment supervises the empirical spectrum against analytic targets for MRW, fGn, and Gaussian controls, using masks for heavy-tailed cases where high-order moments are not reliable. This stage successfully reduces false curvature for fGn and Gaussian controls, but it can also over-linearize genuine MRW curvature. A later curvature-preserving alignment stage adds stronger band-specific curvature losses for medium/high- λ^2 MRW and uses third-difference smoothness to discourage jagged spectra without directly flattening quadratic curvature.

These stages are important as ablations, but they are not presented as a final solution to short-window MRW parameter recovery. Their main value is diagnostic: they show that suppressing fake curvature and preserving real MRW curvature are competing requirements under finite samples.

3.5 Analytic spectrum geometry

Figure 3 illustrates the analytic geometry implied by the projection families. When $\lambda^2 = 0$, the MRW curve reduces to the linear monofractal spectrum. When λ^2 is small, the curve is close to linear and should be treated as a boundary case. When λ^2 is larger, the curvature becomes visible in $\zeta(q)$, in the slope $\alpha(q)$, and in the corresponding $f(\alpha)$ geometry.

3.6 Spectral geometry diagnostics

The final diagnostic vector combines projection and stability quantities:

$$\left(r_{\text{MRW}}, r_{\text{mono}}, \hat{\lambda}_{\text{proj}}^2, \text{gain}_{\text{MRW-vs-mono}}, \text{curvature}, \text{linearity}, \text{boundary}, \text{tail/instability} \right).$$

The diagnostic scores are calibrated from these features. The curvature diagnostics and $\hat{\lambda}_{\text{proj}}^2$ are related but not identical: the former summarize the observed geometry of $\hat{\zeta}(q)$, whereas the latter is the coordinate obtained by projecting that geometry onto the MRW family. These quantities should not be reported in isolation. A useful reading is that p_{MRW} is meaningful only when scaling is stable, curvature is present, the MRW residual is lower than the monofractal residual, and instability warnings are not dominant. This conditional interpretation is the central design principle of CMIN-SR.

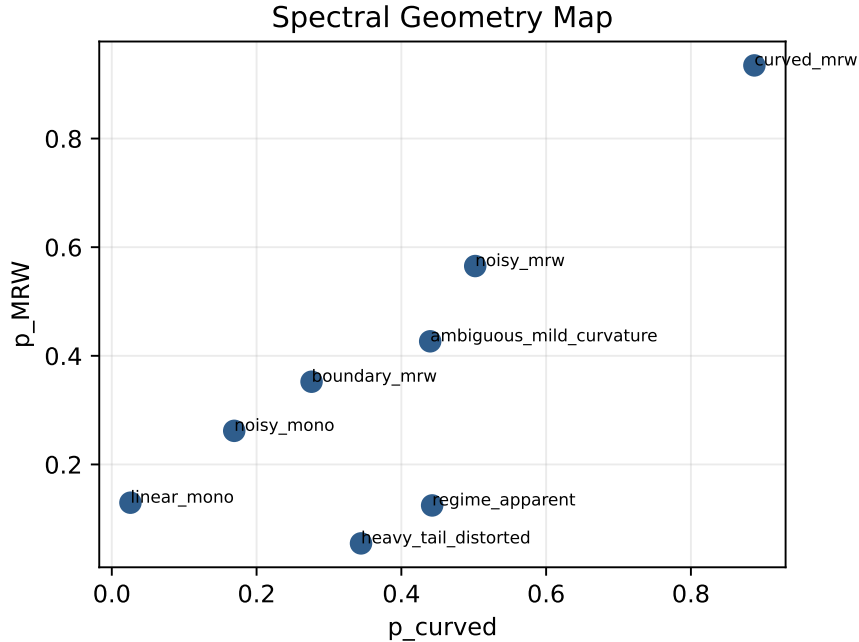


Figure 4: Analytic spectral geometry calibration. Clean monofractal, boundary-MRW, and curved-MRW spectra occupy distinct regions of calibrated spectral diagnostic space.

4 Experimental Design

4.1 Questions tested

The experiments are organized around five questions.

First, if the input spectrum is clean, can the spectral-geometry calibrator distinguish linear monofractal, boundary MRW, curved MRW, and unstable spectra? Second, on finite raw samples, do monofractal and non-MRW controls create false MRW-compatible evidence? Third, can deterministic estimators recover the MRW curvature coordinate λ^2 under the tested sample lengths and scale ranges? Fourth, how much zeta-space noise is needed to destroy the analytic separation between linear and curved spectra? Fifth, can the framework be run on real complex-system data as a cautious diagnostic sanity check?

This ordering is deliberate. The paper does not start from the assumption that MRW parameters are recoverable. It first checks the analytic spectral geometry and then tests where this geometry breaks down under finite-sample estimation.

4.2 Synthetic process families

The controlled raw-sample experiments use the following process families: MRW, low- λ^2 MRW, shuffled MRW, fractional Gaussian noise, iid Gaussian, iid Student- t , GARCH(1,1), and regime-switching Gaussian. The MRW and fGn families provide interpretable scaling targets. Gaussian, Student t , GARCH, and regime-switching controls are included because heavy tails, volatility clustering, and nonstationary variance can produce apparent multiscale behavior without implying an MRW mechanism [7, 17].

The default diagnostic q -grid is $\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$. Process-family diagnostics use lengths

$T = 256, 512, 1024, 2048$. The deterministic identifiability supplement reports short-window grids including $T = 512, 1024, 2048$, with scale sets based on powers of two. All such claims are restricted to the tested grids.

4.3 Analytic spectrum-space experiments

The analytic spectrum-space dataset does not generate raw time series. It directly generates $\zeta(q)$ curves from known families: linear monofractal, boundary MRW, curved MRW, noisy monofractal, noisy MRW, heavy-tail distorted, regime-apparent, and ambiguous mild-curvature spectra. This experiment tests only the interpretation layer. It answers whether linear, boundary, and curved geometries are separable once the spectrum is given.

To check that the analytic result is not seed-specific, the evaluation was repeated for seeds 2024, 2025, and 2026. The resulting mean and standard deviation tables are included in the paper assets. Training configurations, output tables, and figure data are kept in the repository so that the reported diagnostics can be regenerated without relying on undocumented manual processing.

4.4 Deterministic and classical baselines

The finite-sample identifiability study compares several deterministic structure-function variants: ordinary least squares, trimmed regression, bootstrap over scales, and smoothed $\zeta(q)$. In addition, classical aggregate-increment structure functions, first- and second-order MFDFA, and dependency-free Haar wavelet-leader / WTMM-style modulus-maxima baselines were run on the same quick finite-sample grid. These baselines are not used to claim that CMIN-SR outperforms all classical methods. They are used to test whether representative classical estimators remove the short-window λ^2 recovery bottleneck in the tested settings [26, 19, 18, 28].

The Haar wavelet-leader and WTMM-style rows should be interpreted as compact classical controls, not as full production wavelet-leader or continuous-WTMM packages. They are included to test whether lightweight classical alternatives remove the observed bottleneck, not to close the broader benchmarking question.

4.5 Zeta-noise bridge and failure attribution

The zeta-noise bridge adds controlled perturbations to analytic spectra before feeding them to the spectral-geometry calibrator. Smooth, high- q , jagged, and high- q -biased perturbations are included. This experiment measures how the calibrated MRW-vs-linear separation margin decreases as the input spectrum becomes less reliable.

The failure-attribution experiment compares three levels for the same process families: analytic target spectra, deterministic estimated spectra, and neural empirical spectra. This directly tests whether the bottleneck lies in the spectral interpretation layer or in the upstream empirical spectrum.

4.6 Real-world sanity check

As an exploratory real-data check, the diagnostic proxy pipeline is applied to local Fama-French factor returns. The four selected factor series are Mkt-RF, SMB, HML, and Mom. Rolling windows of length 512 are compared against shuffled surrogates. The goal is modest: to verify that the diagnostic shows original-vs-shuffled differences consistent with temporal multiscale dependence. These results are not used to claim that financial factor returns are generated by an MRW process.

Table 1: Process-family diagnostics at $T = 1024$. Scores should be read jointly; high scaling alone is not MRW evidence.

Process	Band	p_c	p_{MRW}	p_{mono}	p_b
MRW	high	0.186	0.306	0.827	0.146
Low- λ^2 MRW	low	0.177	0.301	0.836	0.123
fGn	non-MRW	0.191	0.310	0.824	0.149
iid Gaussian	non-MRW	0.032	0.131	0.962	0.089
iid Student-t	non-MRW	0.370	0.184	0.509	0.062
GARCH(1,1)	non-MRW	0.036	0.136	0.958	0.087
Regime-switching Gaussian	non-MRW	0.333	0.229	0.582	0.100

4.7 Evaluation metrics

The main reported quantities are diagnostic scores ($p_{\text{scaling}}, p_{\text{curved}}, p_{\text{mono}}, p_{\text{MRW}}, p_{\text{boundary}}$), projection residuals, MRW-vs-monofractal gain, λ^2_{proj} , zeta error, and finite-sample recovery summaries such as λ^2 MAE, RMSE, correlation, Spearman correlation, and high- λ^2 detection rate. For real-data sanity checks, the reported quantities are original-minus-shuffled gaps in proxy λ^2 , MRW diagnostic score, spectrum width, and log-volatility covariance slope.

5 Results

5.1 Analytic spectral geometry is stable and separable

The clearest positive result is obtained in analytic spectrum space. Linear monofractal spectra are assigned low curvature and low MRW compatibility, boundary MRW spectra are assigned high boundary score and intermediate MRW compatibility, and strongly curved MRW spectra are assigned high curvature and high MRW compatibility. In the representative evaluation, linear monofractal spectra have $p_{\text{curved}} \approx 0.026$, $p_{\text{MRW}} \approx 0.130$, and $p_{\text{mono}} \approx 0.969$. Curved MRW spectra have $p_{\text{curved}} \approx 0.888$, $p_{\text{MRW}} \approx 0.935$, and $p_{\text{mono}} \approx 0.178$.

The three-seed supplement gives the same qualitative conclusion. Across seeds 2024, 2025, and 2026, curved MRW spectra have mean $p_{\text{curved}} \approx 0.893$, while linear monofractal spectra have mean $p_{\text{curved}} \approx 0.025$. This supports the claim that the spectral interpretation layer learns the intended analytic geometry when the input spectrum is controlled.

5.2 Process-family controls show why diagnostics must be joint

Raw process-family diagnostics show that no single score is sufficient. Monofractal controls can have stable scaling; heavy-tailed and regime-like controls can generate apparent curvature or instability; and GARCH-like volatility clustering produces ambiguous behavior. These controls justify the separation of scaling, curvature, monofractal compatibility, MRW compatibility, boundary behavior, and instability.

The important point is not that the diagnostic labels every process perfectly. The point is that it avoids a single “stable scaling implies MRW” rule. A sample with high scaling evidence but low curvature evidence should not be read as a strong MRW case. A sample with apparent curvature but high instability should also be treated cautiously.

Table 2: Projection diagnostics at $T = 1024$. Residuals and instability contextualize MRW-family compatibility.

Process	λ_p^2	r_{MRW}	r_{mono}	Gain	Instab.
MRW	0.048	0.038	0.051	0.247	0.012
Low-lambda2 MRW	0.050	0.041	0.055	0.259	0.012
fGn	0.045	0.038	0.050	0.231	0.012
iid Gaussian	0.000	0.051	0.045	0.000	0.001
iid Student-t	0.000	0.376	0.405	0.071	0.991
GARCH(1,1)	0.000	0.335	0.370	0.088	0.907
Regime-switching Gaussian	0.000	0.353	0.383	0.078	0.966

Table 3: Classical multifractal baseline summary over the quick finite-sample grid. Wavelet rows use compact dependency-free Haar approximations and should be read as controls, not full production WTMM or wavelet-leader packages.

Estimator	MAE	Corr.	Detect	Boundary
MF DFA	0.069	-0.050	0.717	0.596
MF DFA quadratic	0.071	-0.096	0.067	0.267
Aggregated structure function	0.078	0.013	0.017	0.258
Haar wavelet leader	0.082	-0.014	0.000	0.250
Haar WTMM-style maxima	0.082	–	0.000	0.250

5.3 Classical baselines also show finite-sample spectral error

The supplemental classical baseline pass provides a useful reference point. On the same quick finite-sample grid, first-order MF DFA has the lowest average λ^2 MAE among the added classical baselines and detects many high- λ^2 samples, but its λ^2 correlation remains weak and changes sign across sample lengths. Aggregated structure functions and compact Haar wavelet-leader controls show near-zero or unstable correlation, while the Haar WTMM-style maxima baseline degenerates to a low-curvature estimate on this grid. These results strengthen the finite-sample conclusion rather than showing that a single classical estimator solves short-window curvature recovery.

Historical neural baselines and constrained decoder variants are summarized in the supplementary tables. They are useful ablation evidence, but they are not used to claim a universal neural advantage over classical multifractal estimators.

5.4 Finite-sample curvature recovery is weak in the tested grid

The estimator-level study gives the central negative result. Across the tested estimators, λ_{proj}^2 is only weakly related to the known synthetic MRW curvature coordinate under short windows. In the three-seed quick supplement, the mean λ^2 correlation is close to zero across $T = 512, 1024, 2048$, with estimator-dependent variation but no stable recovery pattern. The high- λ^2 detection rate remains weak in the tested grid.

This result should be read carefully. It does not imply that MRW curvature is never identifiable. It says that, under the current q-grid, scale range, sample lengths, and estimator family, short-window empirical spectra do not reliably preserve the curvature coordinate. This is the main reason the paper frames CMIN-SR as a diagnostic framework rather than as a guaranteed λ^2 recovery engine.

Table 4: Compact finite-sample λ^2 recovery summary. Full estimator-level results are retained in the paper assets.

T	Mean corr.	Best corr.	Best est.	Worst corr.	Detect
512	-0.162	-0.125	ols	-0.177	0.000
1024	0.097	0.137	smoothed	-0.022	0.000
2048	-0.051	-0.051	bootstrap	-0.053	0.000

Finite-sample recovery of the MRW curvature coordinate

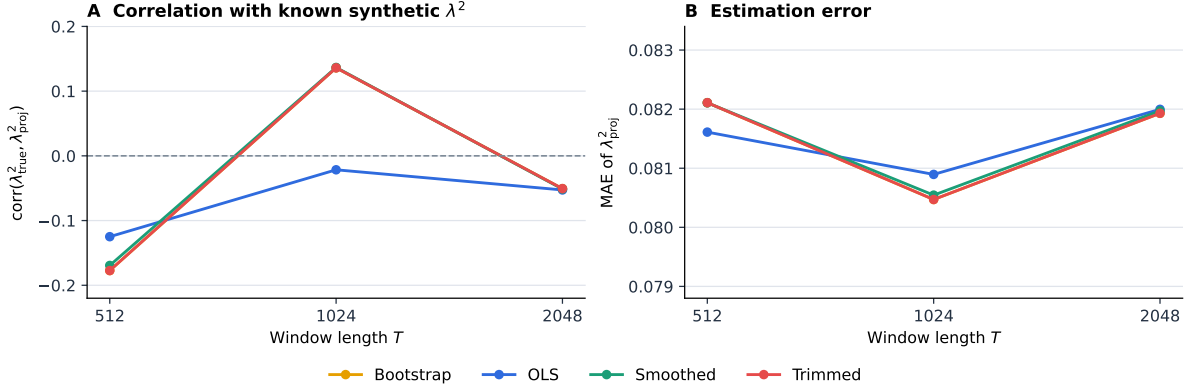


Figure 5: Finite-sample λ^2 recovery for deterministic structure-function estimators. Across the tested short sample lengths, projected λ^2 remains weakly and inconsistently correlated with the known synthetic MRW curvature coordinate, while estimation error stays nearly flat.

5.5 Zeta noise explains the raw-spectrum bottleneck

The zeta-noise bridge shows that spectrum-space separation is sensitive to input-spectrum noise. The MRW-vs-linear p_{MRW} separation margin is about 0.804 at zero noise in the three-seed supplement and decreases to about 0.365 at noise level 0.1. The single refreshed run gives the same qualitative pattern, with the margin falling from about 0.805 to about 0.338. This is a direct explanation of why a successful analytic calibrator can fail when its input $\hat{\zeta}(q)$ is distorted by finite-sample estimation noise.

5.6 Failure attribution localizes the bottleneck

The failure-attribution study compares analytic, deterministic, and neural spectra for the same process families. For high- λ^2 MRW, the analytic spectrum is correctly interpreted as MRW-compatible, with $p_{\text{MRW}} \approx 0.93$. The deterministic structure-function spectrum, however, is interpreted as much less MRW-compatible, with $p_{\text{MRW}} \approx 0.13$ in the refreshed quick run. The neural empirical spectrum improves the zeta MAE relative to the deterministic estimate but still does not restore the analytic MRW curvature, giving $p_{\text{MRW}} \approx 0.33$.

This comparison is important because it separates a failure of spectral interpretation from a failure of empirical spectrum estimation. The calibrator recognizes clean MRW geometry. The difficulty is that short raw samples do not reliably deliver that geometry to the calibrator.

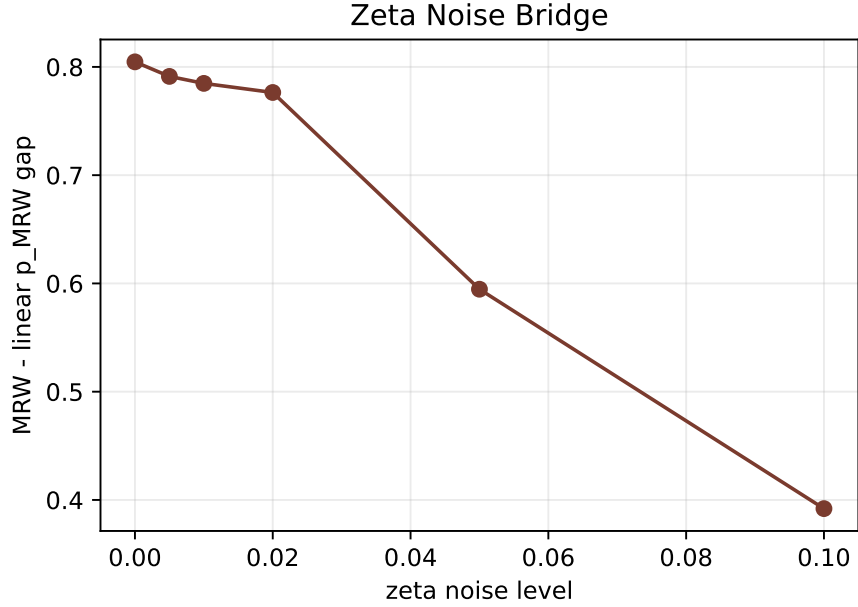


Figure 6: Zeta noise bridge. Increasing q-space noise reduces the separation between linear spectra and curved MRW spectra in calibrated MRW compatibility.

Table 5: Compact ablation summary at $T = 1024$. Values are MRW-compatibility diagnostics, not mechanism probabilities.

Version	MRW	fGn	Gaussian	Student-t
v1	0.906	0.916	0.335	0.108
v2	0.677	0.691	0.365	0.090
v3	0.656	0.639	0.370	0.360
calibrated	0.629	0.638	0.320	0.304

5.7 Ablations clarify the final interpretation

The historical CMIN-SR sequence is best understood as a failure-analysis path, not as a simple line of increasingly better models. Early versions over-associated stable scaling with MRW compatibility. Later versions added monofractal competition, explicit curvature diagnostics, boundary calibration, and raw zeta alignment. These stages improved parts of the diagnostic behavior but also exposed trade-offs. Raw zeta alignment reduced fGn and Gaussian false MRW compatibility, but it also suppressed curvature in generated MRW samples. Curvature-preserving alignment did not fully recover the lost curvature.

The ablation sequence therefore supports the final conclusion: the limiting factor is empirical spectrum quality and finite-sample curvature identifiability, not the absence of another validity head.

5.8 Real-data sanity check

As a cautious application check, the proxy diagnostic was applied to local Fama-French factor returns. Original rolling windows were compared with fully shuffled and block-shuffled surrogates

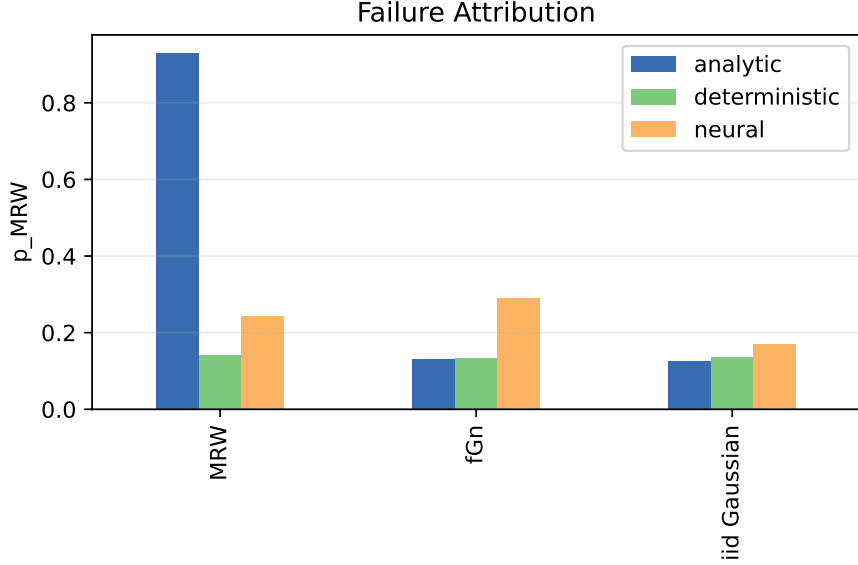


Figure 7: Failure attribution across analytic, deterministic, and neural spectra. Analytic MRW spectra are recognized as MRW-compatible, while deterministic and neural empirical spectra substantially under-recover MRW-family curvature in the tested finite-sample setting.

for Mkt-RF, SMB, HML, and Mom. Under full shuffling, the original-minus-surrogate proxy λ^2 gaps are small but positive: approximately 0.0086 for Mkt-RF, 0.0129 for SMB, 0.0100 for HML, and 0.0137 for Mom. The corresponding MRW-diagnostic gaps are also small and positive. Under block shuffling, the gaps are smaller, near zero, or negative, indicating that the response is sensitive to how much temporal structure the surrogate preserves.

These results are included only as a sanity check. They show that the diagnostic responds to temporal structure that is partially destroyed by shuffling. They do not demonstrate that financial factor returns are generated by an MRW process.

6 Failure Analysis

6.1 Why analytic success does not imply raw-sample success

The analytic spectrum-space experiment deliberately removes the hardest part of the problem: estimating $\zeta(q)$ from a finite sample. Its success shows that the spectral geometry is learnable. It does not show that the same geometry will be visible in a short raw time series. The failure-attribution experiment closes this gap by comparing analytic target spectra, deterministic estimated spectra, and neural empirical spectra for the same process families. The observed failure occurs primarily between the raw sample and the empirical spectrum.

This distinction is important for interpretation. If the analytic calibrator failed, the diagnostic design would be questionable. If deterministic estimators succeeded but the neural encoder failed, the issue would be neural training. Instead, deterministic and neural empirical spectra both struggle to preserve MRW curvature under the tested short-window conditions. The evidence therefore points to finite-sample spectral identifiability as the bottleneck.

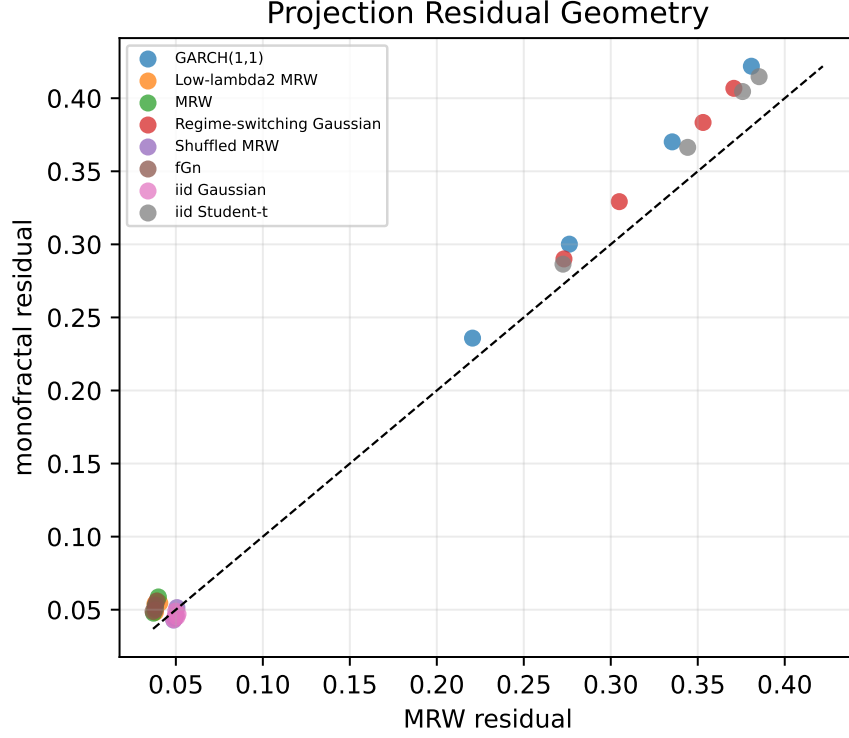


Figure 8: Projection residual geometry. MRW compatibility is interpreted relative to the competing monofractal residual, rather than from scaling stability alone.

6.2 Fake curvature and true curvature are coupled

Raw zeta alignment shows a useful but incomplete success: it suppresses fake curvature in fGn and Gaussian controls. This reduces false MRW compatibility. However, the same smoothing and alignment pressure can suppress true MRW curvature. Curvature-preserving alignment was introduced to counter this effect, using band-specific MRW curvature losses and third-difference smoothness. Even then, medium/high- λ^2 MRW curvature was not reliably restored.

This trade-off is not just an implementation detail. In short samples, small quadratic curvature and finite-sample noise can be comparable in magnitude. Suppressing noise risks flattening real curvature; preserving curvature risks letting monofractal noise appear multifractal. This is precisely why the final framework reports boundary and instability diagnostics instead of forcing a hard MRW/non-MRW decision.

6.3 What the negative result means

The negative identifiability result should not be read as a failure of multifractal analysis in general. It is a statement about the tested finite regime: short windows, limited scale support, the chosen q -grid, and the implemented structure-function estimator family. Larger samples, richer scale ranges, or alternative estimators such as wavelet leaders may improve recovery in some settings. The present result says that, under the conditions studied here, λ_{proj}^2 is not a reliable short-window estimate of the true MRW intermittency parameter.

The practical implication is conservative. CMIN-SR can organize evidence about scaling, curva-

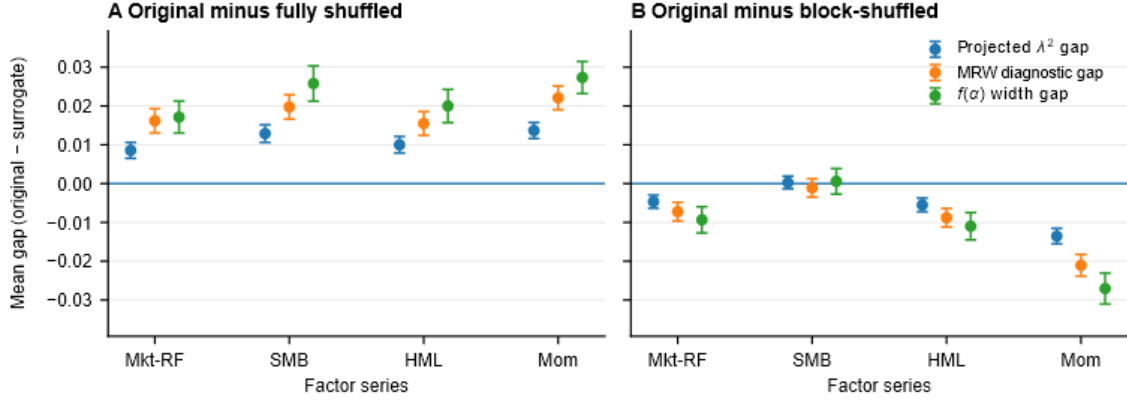


Figure 9: Exploratory real-data sanity check on Fama-French factor returns. Original rolling windows are compared with fully shuffled and block-shuffled surrogates. The gaps are small and should be interpreted as diagnostic responses to temporal multiscale structure, not as MRW mechanism validation.

ture, projection residuals, and instability. It should not be used to announce an MRW mechanism merely because a finite spectrum can be fit by the MRW family.

7 Discussion

7.1 A diagnostic rather than a mechanism test

The main outcome of this study is a diagnostic decomposition of finite-sample multifractal evidence. A high scaling score means that scaling is stable over the tested scales. It does not imply nonlinear curvature. A nonzero λ_{proj}^2 means that the empirical spectrum has an MRW-family projection coordinate. It does not prove that the sample was generated by an MRW. A high MRW diagnostic score should be interpreted only together with the monofractal residual, MRW residual, curvature score, boundary score, and instability warnings.

This caution is not a weakness of the framework; it is the point of the framework. Finite empirical spectra often invite overinterpretation. CMIN-SR is designed to make such overinterpretation harder by forcing the analyst to compare competing explanations.

7.2 Relation to classical multifractal estimators

Classical estimators such as structure functions, MFDFA, and wavelet-based methods remain central to multifractal analysis [25, 26, 19, 18, 28]. CMIN-SR is not proposed as a replacement for these methods. It is better understood as a spectral diagnostic layer that can sit on top of empirical spectra and ask how the estimated geometry relates to monofractal and MRW-family projections.

The supplemental baseline results are consistent with this interpretation. MFDFA can reduce error relative to a simple structure-function baseline in some settings, but high- λ^2 MRW curvature remains difficult in the finite ensemble baseline. The added Haar wavelet-leader and WTMM-style controls provide a lightweight wavelet comparison and do not remove the short-window recovery limitation. A full production WTMM or wavelet-leader package would still be valuable as future benchmarking, but the present classical controls already support the conservative finite-sample conclusion.

7.3 Implications for complex systems

The framework is especially relevant for complex systems where multiple mechanisms can produce similar spectral signatures. In finance, for example, heavy tails, volatility clustering, regime changes, and temporal dependence can all affect scaling statistics [12, 8, 29, 6]. The Fama-French sanity check shows small positive original-vs-shuffled gaps, which is consistent with some temporal multiscale dependence. But this is not MRW mechanism validation. The right conclusion is narrower: the diagnostic can be applied to real return series and shows original-vs-shuffled differences consistent with temporal structure.

7.4 Limitations

Several limitations are important.

First, the framework depends on the MRW parabolic projection family. This makes the diagnostics interpretable, but it also means that non-MRW multifractal processes may be only partially represented. The coordinate λ_{proj}^2 should therefore be interpreted as an MRW-family curvature coordinate, not as a universal intermittency parameter.

Second, the finite-sample identifiability result is conditional on the tested sample lengths, q -grid, scale sets, and estimator family. The current evidence does not rule out better recovery with longer samples, denser scale support, or different estimators. It does show that short-window recovery is not reliable under the tested setting.

Third, the reported diagnostic scores are calibrated scores in the modeling sense, not fully validated statistical probabilities. The project does not yet include reliability diagrams, expected calibration error, or Brier-score calibration analysis. The scores should therefore be read as comparable diagnostic indices rather than posterior probabilities of a data-generating mechanism.

Fourth, the real-data evidence is exploratory. The Fama-French experiment uses surrogate comparison to check diagnostic responsiveness, not to establish a true data-generating mechanism. Stronger real-world claims would require domain-specific preprocessing, additional surrogate designs, uncertainty analysis, and comparison against alternative multifractal estimators.

Finally, the classical baseline comparison is still not exhaustive. Structure functions, MF DFA, and compact Haar wavelet controls are included, but these wavelet rows are not substitutes for a full production WTMM or wavelet-leader benchmark. This should be acknowledged openly in any submission.

8 Conclusion

This paper frames MRW-like inference from finite stochastic signals as a validity-aware spectral diagnostic problem. CMIN-SR separates empirical spectrum estimation from spectral interpretation, compares monofractal and MRW projection families, and reports diagnostic scores for scaling, curvature, monofractal compatibility, MRW compatibility, boundary behavior, and instability.

The experiments support a balanced conclusion. In analytic spectrum space, the MRW and monofractal geometries are separable and stable across random seeds. In raw finite samples, however, empirical spectrum estimation is the limiting step: deterministic and neural zeta estimators do not reliably preserve the MRW curvature coordinate under the tested short-window settings. The added classical baseline comparison strengthens this interpretation: MF DFA improves some metrics relative to simple structure functions, but MF DFA, aggregate structure functions, and lightweight Haar wavelet controls do not produce stable short-window λ^2 recovery on the tested

grid. The real-data sanity check further shows that the framework can be applied to complex financial factor returns, but only as a diagnostic tool, not as a mechanism test.

CMIN-SR should therefore be used to organize evidence for or against MRW-compatible spectral geometry. It should not be used as a guaranteed short-window λ^2 recovery engine or as proof that an observed signal was generated by an MRW process.

References

- [1] Alain Arneodo, Emmanuel Bacry, and Jean-François Muzy. The thermodynamics of fractals revisited with wavelets. *Physica A: Statistical Mechanics and its Applications*, 213(1–2):232–275, 1995.
- [2] Simon Arridge, Peter Maass, Ozan Öktem, and Carola-Bibiane Schönlieb. Solving inverse problems using data-driven models. *Acta Numerica*, 28:1–174, 2019.
- [3] Emmanuel Bacry, Jean Delour, and Jean-François Muzy. Multifractal random walk. *Physical Review E*, 64(2):026103, 2001.
- [4] Emmanuel Bacry, Arnaud Gloter, Marc Hoffmann, and Jean-François Muzy. Multifractal analysis in a mixed asymptotic framework. *The Annals of Applied Probability*, 20(3):1101–1146, 2010.
- [5] Emmanuel Bacry and Jean-François Muzy. Log-infinitely divisible multifractal processes. *Communications in Mathematical Physics*, 236:449–475, 2003.
- [6] Jozef Barunik, Tomaso Aste, Tiziana Di Matteo, and Ruipeng Liu. Understanding the source of multifractality in financial markets. *Physica A: Statistical Mechanics and its Applications*, 391(17):4234–4251, 2012.
- [7] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.
- [8] Jean-Philippe Bouchaud, Marc Potters, and Martin Meyer. Apparent multifractality in financial time series. *The European Physical Journal B*, 13(3):595–599, 2000.
- [9] Steven L. Brunton, Bernd R. Noack, and Petros Koumoutsakos. Machine learning for fluid mechanics. *Annual Review of Fluid Mechanics*, 52:477–508, 2020.
- [10] Laurent E. Calvet and Adlai J. Fisher. Multifractality in asset returns: Theory and evidence. *The Review of Economics and Statistics*, 84(3):381–406, 2002.
- [11] Ashvin Chhabra and Roderick V. Jensen. Direct determination of the $f(\alpha)$ singularity spectrum. *Physical Review Letters*, 62(12):1327–1330, 1989.
- [12] Rama Cont. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001.
- [13] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- [14] Tiziana Di Matteo, Tomaso Aste, and Michel M. Dacorogna. Long-term memories of developed and emerging markets: Using the scaling analysis to characterize their stage of development. *Journal of Banking & Finance*, 29(4):827–851, 2005.

- [15] Dariusz Grech and Grzegorz Pamuła. On the multifractal effects generated by monofractal signals. *Physica A: Statistical Mechanics and its Applications*, 392(23):5845–5864, 2013.
- [16] Thomas C. Halsey, Mogens H. Jensen, Leo P. Kadanoff, Itamar Procaccia, and Boris I. Shraiman. Fractal measures and their singularities: The characterization of strange sets. *Physical Review A*, 33(2):1141–1151, 1986.
- [17] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.
- [18] Stéphane Jaffard, Bruno Lashermes, and Patrice Abry. Wavelet leaders in multifractal analysis. In *Wavelet Analysis and Applications*, pages 201–246. Birkhäuser, 2007.
- [19] Jan W. Kantelhardt, Stephan A. Zschiegner, Eva Koscielny-Bunde, Shlomo Havlin, Armin Bunde, and H. Eugene Stanley. Multifractal detrended fluctuation analysis of nonstationary time series. *Physica A: Statistical Mechanics and its Applications*, 316(1–4):87–114, 2002.
- [20] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021.
- [21] Josef Ludescher, Armin Bunde, Christian L. E. Franzke, and Hans Joachim Schellnhuber. Finite-size effects in multifractal detrended fluctuation analysis. *Physica A: Statistical Mechanics and its Applications*, 390(13):2480–2490, 2011.
- [22] Stephane Mallat and Wen Liang Hwang. Singularity detection and processing with wavelets. *IEEE Transactions on Information Theory*, 38(2):617–643, 1992.
- [23] Benoit B. Mandelbrot, Adlai J. Fisher, and Laurent E. Calvet. A multifractal model of asset returns. Technical Report 1164, Cowles Foundation for Research in Economics, Yale University, 1997.
- [24] Benoit B. Mandelbrot and John W. Van Ness. Fractional brownian motions, fractional noises and applications. *SIAM Review*, 10(4):422–437, 1968.
- [25] Jean-François Muzy, Emmanuel Bacry, and Alain Arneodo. Wavelets and multifractal formalism for singular signals: Application to turbulence data. *Physical Review Letters*, 67(25):3515–3518, 1991.
- [26] Jean-François Muzy, Emmanuel Bacry, and Alain Arneodo. Multifractal formalism for fractal signals: The structure-function approach versus the wavelet-transform modulus-maxima method. *Physical Review E*, 47(2):875–884, 1993.
- [27] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [28] Herwig Wendt and Patrice Abry. Bootstrap for empirical multifractal analysis. *IEEE Signal Processing Magazine*, 24(4):38–48, 2007.
- [29] Wei-Xing Zhou. The components of empirical multifractality in financial returns. *EPL (Europhysics Letters)*, 88(2):28004, 2009.

A Ablation and Legacy Experiment Notes

Recommended appendix organization:

- Tiny CMIN: always-MRW bias baseline.
- CMIN-Robust: anti-confounding with heavy-tail, shuffled, GARCH, and regime controls.
- CMIN-SR v1/v2/v3: progression from scaling-only to monofractal competition and explicit curvature diagnostics.
- Boundary calibration: same-H raw contrastive calibration was insufficient.
- Raw zeta alignment: suppresses fake fGn/Gaussian curvature but over-linearizes MRW.
- Curvature-preserving zeta alignment: confirms the trade-off between fGn rejection and MRW curvature recovery.
- Finite-sample identifiability: deterministic estimators also struggle, supporting the limitation claim.

Use `notes/legacy_experiments_map.md` for detailed stage-by-stage descriptions.